CSE 242 - ML ASSIGNMENT - 1

PART 1: Linear Algebra

Q1] Matrices given : $A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

↳ AB

$$= \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3(1) + 4(0) & 3(2) + 4(-1) \\ 2(1) + 1(0) & 2(2) + 1(-1) \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

↳ BA

$$= \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1(3) + 2(2) & 1(4) + 2(1) \\ 0(3) + (-1)(2) & 0(4) + (-1)(1) \end{bmatrix}$$

$$BA = \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$$

↳ $(A+B)^T$

$$A+B = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}$$

$$(A+B)^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$$

↳ AB^T

$$B^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$$

$$AB^T = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3(1)+4(2) & 3(0)+4(-1) \\ 2(1)+1(2) & 2(0)+1(-1) \end{bmatrix}$$

$$\boxed{AB^T = \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}}$$

↳ trace (A)

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$$

trace (A) = 3 + 1

$$= \boxed{4}$$

↳ trace (AB)

$$AB = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

trace (AB) = 3 + 3

$$= \boxed{6}$$

Q2) Given $M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

Calc ↳ Eigenvalue
↳ Eigen vector
↳ Verify soln by showing $Mv = \lambda v$

$\because M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

$$M - \lambda I = \begin{bmatrix} (2-\lambda) & -1 \\ -1 & (2-\lambda) \end{bmatrix}$$

The characteristic eq" is $|M - \lambda I| = 0$

Solving Determinant

$= (2-\lambda)^2 - (-1)(-1) = 0$

$= 4 - 4\lambda + \lambda^2 - 1 = 0$

$= \lambda^2 - 4\lambda + 3 = 0$

OR

$$\boxed{\lambda^2 - \{\text{sum of diagonal}\}\lambda + |M| = 0}$$

$= \lambda^2 - 4\lambda + 3 = 0$

$= \lambda(\lambda-1) - 3(\lambda-1) = 0$

$= (\lambda-3)(\lambda-1) = 0$

$\therefore \lambda = 3$ or $\lambda = 1$ are Roots & Eigen Value

case ① : When $\lambda = 3$

consider $(M - \lambda I)V = 0$

$$(M - 3I)V = 0$$

$$\left[\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right]\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$R_2 - R_1$

$$\begin{bmatrix} -1 & -1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Eq^n is $\boxed{-v_1 - v_2 = 0}$

put $v_2 = t$

$$-v_1 - t = 0$$

$$\boxed{v_1 = -t}$$

$$\therefore \boxed{V_1 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -t \\ t \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} t}$$ Eigen vector

case ② : When $\lambda = 1$

consider $(M - \lambda I)V = 0$

$$(M - I)V = 0$$

$$= \left[\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right]\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$R_2 + R_1$

$$= \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\therefore$ Eq^n is $\boxed{v_1 - v_2 = 0}$

put $v_2 = t$

$v_1 - t = 0$

$\therefore v_1 = t$

$$V_2 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} t$$ Eigen Vector

↳ $MV = \lambda V$

Case ① : When $\lambda = 3$

$$MV_1 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -t \\ t \end{bmatrix}$$

$$= \begin{bmatrix} -2t - t \\ t + 2t \end{bmatrix} = \begin{pmatrix} -3t \\ 3t \end{pmatrix}$$

$$\lambda_1 V_1 = 3 \begin{pmatrix} -t \\ t \end{pmatrix} = \begin{pmatrix} -3t \\ 3t \end{pmatrix}$$

$\therefore MV_1 = \lambda_1 V_1$

Case ② : When $\lambda = 1$

$$MV_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} t \\ t \end{bmatrix}$$

$$= \begin{bmatrix} 2t - t \\ -t + 2t \end{bmatrix} = \begin{pmatrix} t \\ t \end{pmatrix}$$

$$\lambda_2 V_2 = 1 \begin{pmatrix} t \\ t \end{pmatrix} = \begin{pmatrix} t \\ t \end{pmatrix}$$

$\therefore MV_2 = \lambda_2 V_2$

Q3) To prove trace (AB) = trace (BA)

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -5 & -6 \\ -7 & -8 \end{bmatrix}$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} -5 & -6 \\ -7 & -8 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-5) + 2(-7) & 1(-6) + 2(-8) \\ 3(-5) + 4(-7) & 3(-6) + 4(-8) \end{bmatrix}$$

$$AB = \begin{bmatrix} -5 - 14 & -6 - 16 \\ -15 - 28 & -18 - 32 \end{bmatrix} = \begin{bmatrix} -19 & -22 \\ -43 & -50 \end{bmatrix}$$

$$BA = \begin{bmatrix} -5 & -6 \\ -7 & -8 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} -5(1) - 6(3) & -5(2) - 6(4) \\ -7(1) - 8(3) & -7(2) - 8(4) \end{bmatrix}$$

$$BA = \begin{bmatrix} -5 - 18 & -10 - 24 \\ -7 - 24 & -14 - 32 \end{bmatrix} = \begin{bmatrix} -23 & -34 \\ -31 & -46 \end{bmatrix}$$

trace $(AB) = -19 - 50 = -69$

trace $(BA) = -23 - 46 = -69$

$\therefore$ trace (AB) = trace (BA)

Q4] Given $C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

↳ to show that inverse exists & compute it

$$|C| = 4(6) - 7(2)$$

$$= 10$$

As $|C| \neq 0$

$\therefore$ Inverse exists

$$C^{-1} = \frac{1}{|C|} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 6/10 & -7/10 \\ -2/10 & 4/10 \end{bmatrix}$$

$$C^{-1} = \begin{bmatrix} 3/5 & -7/10 \\ -1/5 & 2/5 \end{bmatrix}$$

PART 2: Probability, MLE & MAP Estimation

Q1) Discrete distribution

$$P(X=k) = c \cdot \frac{3^k}{k!}, \quad k = 0, 1, 2, \ldots$$

↳ Find normalisation const c.

Probabilities must sum to 1,

$$\sum_{k=0}^{\infty} P(X=k) = 1$$

$$\therefore \quad c \sum_{k=0}^{\infty} \frac{3^k}{k!} = 1$$

By exponential series

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

Here

$x = 3$

$$\therefore \quad c e^3 = 1$$

$$\therefore \quad \boxed{c = e^{-3}}$$

$\therefore$ distribution is

$$P(X=k) = e^{-3} \frac{3^k}{k!} \quad \text{— (i)}$$

↳ Compute Expectation $E[X]$ & Variance $Var(X)$

By Poisson distribution

$$P(X=x) = e^{-m} \frac{m^x}{x!} \quad \text{— (ii)}$$

By (i) & (ii)

$\therefore m = 3$ where m = mean = variance

$$\therefore \boxed{E(X) = 3 \quad \& \quad Var(X) = 3}$$

Q2) $P(A) = 0.5$, $P(B) = 0.3$, $P(A|B) = 0.6$

↳ $P(A \cap B)$

$= P(A|B) \cdot P(B)$

$= 0.6 \times 0.3$

$= 0.18$

↳ $P(B|A)$

$= \frac{P(A \cap B)}{P(A)}$

$= \frac{0.18}{0.5}$

$= 0.36$

↳ $P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= 0.5 + 0.3 - 0.18$

$= 0.62$

Q3) $P(\text{Heads}) = \theta$ — given

$P(\text{Tails}) = 1 - \theta$

Coin flipped 5 times with sequence – H, H, T, H, T

Heads = 3

Tails = 2

↳ Maximum Likelihood Estimate for θ

Likelihood function $L(\theta) = P(H, H, T, H, T | \theta)$

$= \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta)$

$L(\theta) = \theta^3 \cdot (1-\theta)^2$

Log Likelihood : $l(\theta) = \text{Log } L(\theta)$

$= \log [\theta^3 (1-\theta)^2]$

$= 3 \log \theta + 2 \log (1-\theta)$

— differentiation

$$\frac{dl}{d\theta} = \frac{3}{\theta} - \frac{2}{1-\theta}$$

setting derivative to zero

$$\frac{3}{\theta} - \frac{2}{1-\theta} = 0$$

$$\therefore \frac{3}{\theta} = \frac{2}{1-\theta}$$

$$3(1-\theta) = 2\theta$$
$$3 - 3\theta = 2\theta$$
$$3 = 5\theta$$
$$\theta_{MLE} = \frac{3}{5} = 0.6$$

↳ Maximum a Posterior (MAP) Estimate

$$P(\theta) = \begin{cases} 2\theta & \text{for } 0 \le \theta \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Posterior $\propto$ Likelihood $\propto$ Prior

Multiplying likelihood by prior

$$L(\theta) = \theta^3 (1-\theta)^2$$

$$P(\theta) = 2\theta$$

$$\therefore P(\theta \mid \text{data}) \propto \theta^3 (1-\theta)^2 (2\theta)$$

$$P(\theta \mid \text{data}) \propto 2\theta^4 (1-\theta)^2$$

$$\boxed{P(\theta \mid \text{data}) \propto \theta^4 (1-\theta)^2}$$

— Taking the log

$$l(\theta) = 4 \log \theta + 2 \log (1-\theta) + C$$

— differentiate

$$\frac{dl}{d\theta} = \frac{4}{\theta} - \frac{2}{1-\theta}$$

— set equal to 0

$$\frac{4}{\theta} - \frac{2}{1-\theta} = 0$$

$$\therefore \frac{4}{\theta} = \frac{2}{1-\theta}$$

$$4(1-\theta) = 20$$
$$4 - 4\theta = 20$$
$$4 = 6\theta$$

$$\theta_{MAP} = \frac{4}{6} = \frac{2}{3} = 0.667$$

04) Given : $x = \{1, 3, 5, 7\}$, $n = 4$

μ is mean, σ^2 is variance

$$L(\mu, \sigma^2) = \prod_{i=1}^{4} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

$$\therefore \quad L(\mu, \sigma^2) = (2\pi\sigma^2)^{-4/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{4}(x_i - \mu)^2}$$

Taking log

$$l(\mu, \sigma^2) = -2\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{4}(x_i - \mu)^2$$

differentiate wrt μ

$$\frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{4}(x_i - \mu)$$

$$\frac{1}{\sigma^2}\sum_{i=1}^{4}(x_i - \mu) = 0$$

$$\therefore \quad \sum_{i=1}^{4}(x_i - \mu) = 0$$

Substituting 1, 3, 5, 7:

$$(1-\mu) + (3-\mu) + (5-\mu) + (7-\mu) = 0$$
$$16 - 4\mu = 0$$

$$\mu_{MLE} = 4$$

differentiate wrt σ^2

$$\frac{\partial l}{\partial \sigma^2} = -\frac{2}{\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{4}(x_i - \mu)^2$$

$$-\frac{2}{\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{4}(x_i - \mu)^2 = 0$$

Multiplying by $2(\sigma^2)^2$

$$-4\sigma^2 + \sum_{i=1}^{4} (x_i - \mu)^2 = 0$$

$$\sigma^2 = \frac{1}{4} \sum_{i=1}^{4} (x_i - \mu)^2$$

As $\mu = 4$

$$\therefore \quad \sigma^2 = \frac{1}{4} \left[(1-4)^2 + (3-4)^2 + (5-4)^2 + (7-4)^2 \right]$$

$$= \frac{1}{4} \left[9 + 1 + 1 + 9 \right]$$

$$\boxed{\sigma^2_{MLE} = 5}$$

Q5] Given: $x = 5$, $\sigma^2 = 4$, $\mu \sim N(0, 1)$

$$\mu_0 = 0 \quad , \sigma_0^2 = 1$$

likelihood for $x = 5$ is

$$P(x|\mu) \propto \exp\left[\frac{-(x-\mu)^2}{2(4)} \right]$$

Substituting $x = 5$

$$P(5|\mu) \propto \exp\left[\frac{-(5-\mu)^2}{8} \right]$$

$$\mu \sim N(0, 1)$$

$$\therefore \quad P(\mu) \propto \exp\left[\frac{-(\mu - 0)^2}{2(1)} \right]$$

$$= \exp\left[-\frac{\mu^2}{2} \right]$$

$$P(\mu|x) \propto P(x|\mu) P(\mu)$$

$$\therefore \quad P(\mu|5) \propto \exp\left[\frac{-(5-\mu)^2}{8} \right] \exp\left[-\frac{\mu^2}{2} \right]$$

Combining exponentials

$$P(\mu|5) \propto \exp\left[-\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}\right]$$

log

$$L(\mu) = -\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}$$

differentiating

$$\frac{dL}{d\mu} = \frac{5-\mu}{4} - \mu$$

$$\frac{5-\mu}{4} - \mu = 0$$

Multiply by 4

$$5 - \mu - 4\mu = 0$$

$$5 - 5\mu = 0$$

$$\mu = 1$$

$$\mu_{MAP} = 1$$

Q6) Prove variance expansion identity :—

$$Var(X) = E(X^2) - [E(X)]^2$$

Variance is

$$Var(X) = E[(X - E(X))^2]$$

Let

$$\mu = E[X]$$

$$Var(X) = E[(X - \mu)^2]$$

$$(X - \mu)^2 = X^2 - 2\mu X + \mu^2$$

$$\therefore \quad Var(X) = E[X^2 - 2\mu X + \mu^2]$$

$$E[X^2] - 2\mu E[X] + E(\mu^2)$$

μ is a constant

$$E[\mu^2] = \mu^2$$

Also

$$E[X] = \mu$$

$$\therefore \quad \text{Var}(X) = E[X^2] - 2\mu^2 + \mu^2$$

$$\text{Var}(X) = E[X^2] - \mu^2$$

Since

$$\mu = E[X]$$

$$\boxed{\text{Var}(X) = E[X^2] - [E[X]]^2}$$

PART 3 : PCA

Q1] To derive $\quad \text{Var}(u^T X) = u^T \Sigma u$

Let $\quad X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_d \end{bmatrix}$

be a random vector with mean

$$\mu = E[X]$$

Covariance matrix is defined -

$$\Sigma = E[(X-\mu)(X-\mu)^T]$$

it's $(i,j)^{th}$ element is

$$\Sigma_{ij} = \text{Cov}(X_i, X_j)$$

Now, projection of X onto the vector u is scalar

$$Y = u^T X$$

want to find

$$\text{Var}(Y) = \text{Var}(u^T X)$$

Now,

$$E[Y] = E[u^T X]$$

$$E[Y] = u^T E[X]$$

Since $E[X] = \mu$

$$E[Y] = u^T \mu$$

Now by def^n of variance

$$Var(Y) = E[(Y - E[Y])^2]$$

Substitute

$$Y = u^T X$$

$$E[Y] = u^T \mu$$

$$Var(u^T X) = E[(u^T X - u^T \mu)^2]$$

$$= E[(u^T (X - \mu))^2]$$

For any scalar a

$$a^2 = a a^T$$

$$\therefore (u^T (X-\mu))^2 = u^T (X-\mu)(X-\mu)^T u$$

$$Var(u^T X) = E[u^T (X-\mu)(X-\mu)^T u]$$

$$= u^T E[(X-\mu)(X-\mu)^T] u$$

As $\Sigma = E[(X-\mu)(X-\mu)^T]$

$\therefore$ $\boxed{Var(u^T X) = u^T \Sigma u}$ is the covariance matrix of X